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InvenIQ

Agentic Inventory Intelligence for Malaysian SMEs

Powered by Groq API · Llama 3.3 70B Versatile

UMHackathon 2026 · Domain 2: AI for Economic Empowerment & Decision Intelligence

Problem Statement & Background

15–20%

Potential revenue lost

to dead stock & stockouts annually

97%

of Malaysian businesses

are SMEs — yet most lack affordable data tools

Days

of manual labour required

for a single inventory restock analysis

Revenue Bleeding

Inventory Mismanagement

Malaysian micro-SMEs operate on razor-thin margins. Fragmented operational data across sales, stock and market signals is rarely consolidated into actionable intelligence — resulting in overstocking (capital lock-up) and understocking (lost revenue opportunities).

Owners cannot react to demand shifts driven by weather, festivals, or competitor price changes, repeating costly procurement mistakes every cycle.

No Affordable Tools

Expensive or Ineffective Alternatives

Professional financial analytics solutions (SAP, Oracle ERP) remain too costly and complex for micro-SMEs. Manual spreadsheets are highly error-prone and completely lack predictive or contextual reasoning capability.

Inventory planning today demands extensive labour work and specialised software skills — resources most kedai runcit owners simply don't have.

Gut-Feeling Procurement

Intuition-Based Decisions

Without affordable analytics, SME owners resort entirely to gut-feel for restocking. There is no mechanism to correlate historical sales data with unstructured external signals like monsoon season, public holidays, or news.

This locks owners in a reactive cycle — panic-buying before peak seasons and holding dead stock afterwards — preventing them from competing with larger data-driven corporations.

Strategic Fit (PRD §2): InvenIQ aligns with the AI-Native development lifecycle — evolving the SME owner's role from manual operator to strategic decision curator.

Proposed Solution

Groq · Llama 3.3 70B

Main Goal

InvenIQ is an AI-powered Decision Intelligence partner that transforms raw operational data into actionable economic strategies — democratising high-level financial analytics for SMEs. Powered by Groq's ultra-low-latency inference, the platform delivers explainable recommendations and profit-oriented decision support, emphasising the WHY behind every recommendation to ensure transparency and trust.

Key Functionalities

Automated Forecasting

Predicts demand based on historical sales data and seasonal trends from the uploaded CSV.

Strategic Recommender

Suggests specific reorder points and discount levels to clear slow-moving stock and recover capital.

Natural Language Insights

Explains the "why" behind data trends in plain language — e.g. "Sales dropped due to unseasonably rainy weather" — no technical training needed.

AI Model & Prompt Design

Model Selected

Groq API with Llama 3.3 70B Versatile — chosen for superior reasoning across structured datasets and generating human-readable strategic explanations at ultra-low latency.

Prompting Strategy

Multi-step agentic prompting: first analyse the data → cross-reference with user intent → generate the recommendation with full economic reasoning.

Input Handling

Client-side JS chunking processes oversized CSVs in segments. Max 2,000 rows per analysis to maintain context within the model's token limit.

Fallback Behaviour

If the model returns a hallucinated or off-topic response, the system triggers a graceful error state, asking the user to refine the prompt or re-upload data.

Target Users & User Stories

User Story

"As an SME Owner, I want to enter a business intent in my own words (e.g. "Clear out old stock to make room for new arrivals") so that InvenIQ provides me with an optimised inventory layout without me having to manually calculate stock levels."

SME Owners

Individuals running small retail or service businesses (e.g. Kedai Runcit) who lack formal financial training. Primary decision-makers for procurement — previously relying entirely on intuition.

Retail Managers

Decision-makers responsible for inventory procurement and stock-level management across one or more locations. Need fast, data-backed restock decisions without waiting for manual analysis.

Financial Analysts

Generate rapid insights from multi-source data — structured sales logs combined with unstructured market trends and news URLs — for client advisory and business forecasting reports.

Operations Leads

Focused on reducing overhead costs and improving supply chain efficiency. Use InvenIQ to identify slow-moving stock, optimise reorder timing, and minimise capital locked in dead inventory.

Inventory Clerks

Export daily POS data as CSV → InvenIQ handles the entire analysis pipeline. Dramatically reduces manual counting, spreadsheet errors, and labour hours spent on weekly stock reviews.

Gig Hub Managers

Community-level coordinators managing resource allocation and task scheduling for local worker groups. Use InvenIQ to predict demand surges and allocate resources for peak periods.

Scope, Assumptions & Risks

In Scope

Data Upload Portal

Supports CSV / Excel files for structured sales data ingestion.

Conversational Dashboard

A natural language chat interface for querying data without technical skills.

Agentic JSON UI Cards

AI outputs strict JSON → rendered as colour-coded (Red/Amber/Green) dashboard cards.

Strategy Export

One-click export of recommended reorder lists and economic reasoning to PDF.

Out of Scope

Direct ERP Integration

No live connection to SAP, Oracle or other enterprise POS systems in this version.

Real-time IoT Tracking

Does not support real-time sensor-based inventory monitoring.

User Authentication

No login screens or persistent cloud-saved chat histories in the MVP.

Automated Purchasing

System recommends but does not auto-execute purchase orders with suppliers.

⚠ Assumptions & Risks

Stable Connectivity

Users need a stable internet connection for Groq LLM cloud inference.

Min. 3 Months Data

System requires ≥3 months of historical CSV data for accurate forecasts.

1024 Token Cost Cap

Max token cost per response. Mitigated via client-side JS pre-computation & query caching.

Human-in-the-Loop

Final procurement decisions require human verification — system is advisory only.

Data Accuracy Risk

AI hallucination mitigated via temperature=0.1, Master Prompt grounding & Rule 8 protocol.

Quality Assurance & Closing

QATD — Test Strategy, Risk Assessment & Anti-Hallucination

Shift-Left Testing

Agile iterative QA, not just end-of-sprint

PyTest + Postman

Unit & integration test frameworks used

Rule 3 100% Enforced

Scope breach blocked; professional tone kept

Temp = 0.1

Deterministic AI output; no creative guessing

0 P0/P1 Critical Bugs

No critical defects found in final build

Risk	L × S	Score	Mitigation Strategy
AI Hallucination — inventing products or figures	5 × 4	20 — Critical	temperature=0.1 + Master Prompt grounding + Rule 8 Anti-Hallucination Protocol + Human-in-the-loop UI
LLM Response Latency — Groq inference delay	3 × 5	15 — High	"Typing indicator" <200ms handshake + async FastAPI + graceful degradation on timeout

Anti-Hallucination Protocol

- Temp = 0.1 — deterministic token selection
- Master Prompt grounding — AI asks user if data is missing
- Rule 8 — only recommends Product IDs from the uploaded CSV
- Human-in-the-loop — markdown math visible to user for verification
- Zero-Data Fallback — refuses to project without CSV upload

From gut-feeling to data-driven — in seconds, not spreadsheets.

InvenIQ acts as a virtual CFO for every Malaysian SME owner.